

Task 2 - Synthesis & Gate-Level Simulation

Objective

To synthesize RTL design into a gate-level netlist and verify its functionality.

Theory

Synthesis converts RTL code into logic gates. Gate-level simulation represents real hardware behavior.

Tools Used

-EDA Environment: EDA Playground

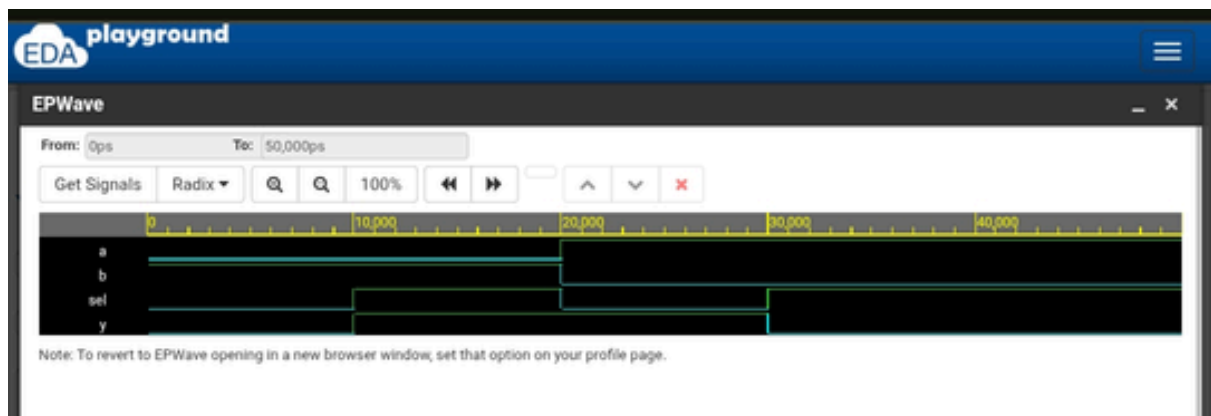
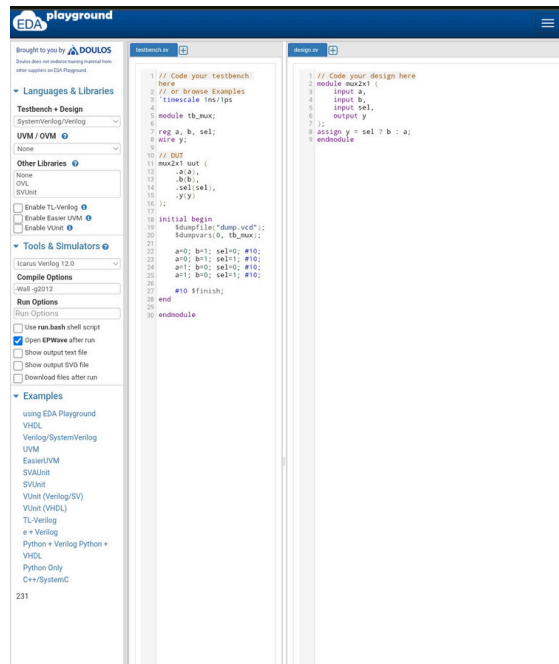
Logic Synthesis: Yosys

Simulation: Icarus Verilog

Design

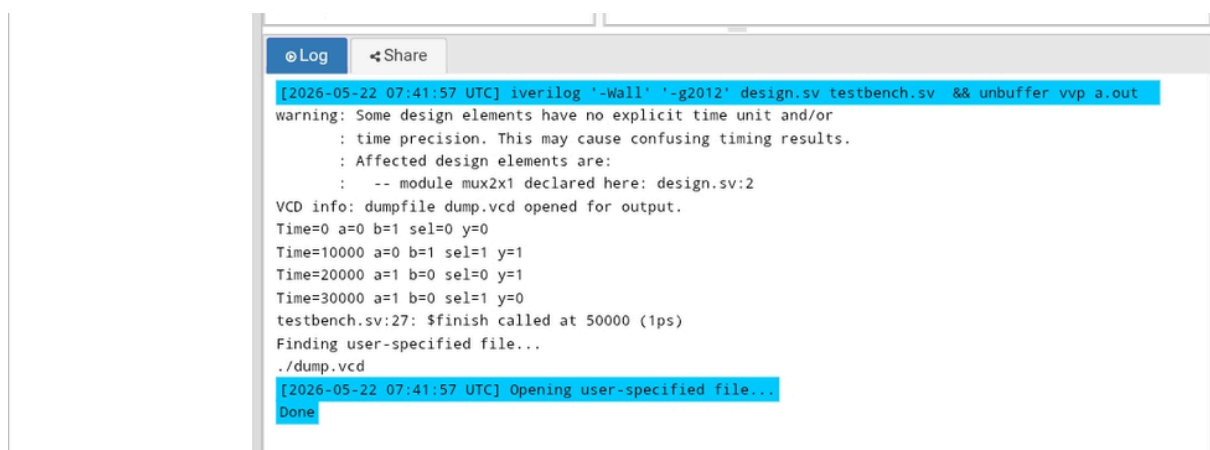
2:1 Multiplexer selects one input based on select signal.

Simulation



RTL Simulation

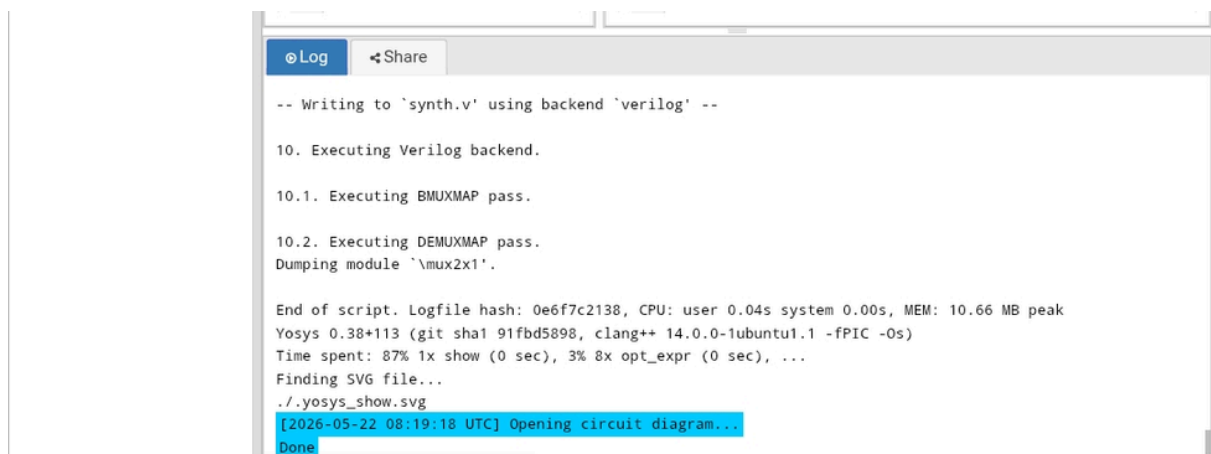
The functional verification of the RTL code was performed using Icarus Verilog. The testbench verified that the multiplexer correctly drives the output y based on the select signal sel.



```
© Log  Share
[2026-05-22 07:41:57 UTC] iverilog '-Wall' '-g2012' design.sv testbench.sv && unbuffer vvp a.out
warning: Some design elements have no explicit time unit and/or
: time precision. This may cause confusing timing results.
: Affected design elements are:
:   -- module mux2x1 declared here: design.sv:2
VCD info: dumpfile dump.vcd opened for output.
Time=0 a=0 b=1 sel=0 y=0
Time=10000 a=0 b=1 sel=1 y=1
Time=20000 a=1 b=0 sel=0 y=1
Time=30000 a=1 b=0 sel=1 y=0
testbench.sv:27: $finish called at 50000 (1ps)
Finding user-specified file...
./dump.vcd
[2026-05-22 07:41:57 UTC] Opening user-specified file...
Done
```

#Synthesis Log & Gate-Level Diagram

The synthesis was successfully completed using Yosys. The log confirms that the design was synthesized into the target technology.



```
Log Share

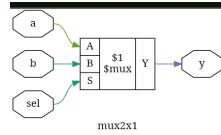
-- Writing to 'synth.v' using backend 'verilog' --

10. Executing Verilog backend.

10.1. Executing BMUXMAP pass.

10.2. Executing DEMUXMAP pass.
Dumping module `mux2x1'.

End of script. Logfile hash: 0e6f7c2138, CPU: user 0.04s system 0.00s, MEM: 10.66 MB peak
Yosys 0.38+113 (git sha1 91fbd5898, clang++ 14.0.0-1ubuntu1.1 -fPIC -Os)
Time spent: 87% 1x show (0 sec), 3% 8x opt_expr (0 sec), ...
Finding SVG file...
./yosys_show.svg
[2026-05-22 08:19:18 UTC] Opening circuit diagram...
Done
```



Comparison

Feature	RTL	Gate-Level
---------	-----	------------

Output	Same	Same
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Delay	No	Yes
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Note: While both simulations confirmed the logical correctness of the design, the gate-level simulation provides a structural netlist

that is closer to actual hardware implementation.

 Files

- mux.v

```
module mux2x1 (
```

```
    input a,
```

```
    input b,
```

```
    input sel,
```

```
    output y
```

```
);
```

```
assign y = sel ? b : a;
```

```
endmodule
```

```
- tb_mux.v
```

```
`timescale 1ns/1ps
```

```
module tb_mux;
```

```
reg a, b, sel;
```

```
wire y;
```

```
// DUT
```



```
mux2x1 uut (
```

```
    .a(a),
```

```
    .b(b),
```

```
    .sel(sel),
```

```
    .y(y)
```

```
);
```

```
initial begin
```

```
    $dumpfile("dump.vcd");
```

```
    $dumpvars(0, tb_mux);
```

```
    $monitor("Time=%0t a=%b
```

```
b=%b sel=%b y=%b", $time, a,  
b, sel, y);
```

```
a=0; b=1; sel=0; #10;
```

```
a=0; b=1; sel=1; #10;
```

```
a=1; b=0; sel=0; #10;
```

```
a=1; b=0; sel=1; #10;
```

```
#10 $finish;
```

```
end
```

```
endmodule
```

Conclusion

The synthesis of the 2:1 multiplexer was successful. The gate-level netlist and the gate diagram confirm that the tool effectively mapped the HDL code into structural components. Both RTL and gate-level simulations performed as expected, confirming design integrity.